

Cyclistic Bike-share Case Study

1. Business task

The purpose of this analysis is to understand how annual members and casual riders use Cyclistic bikes differently in order to support the development of marketing strategies that could convert casual riders into annual members.

2. Data sources

The dataset used for this project consists of historical Cyclistic trip data covering previous 12 months, from May 2025 to April 2026. The data was downloaded on May 24, 2026, from Motivate International Inc, in CSV format.

The dataset was organized into twelve separate monthly files.

To ensure data integrity:

- Raw files were preserved separately from the cleaned dataset
- Duplicate detection checks were performed
- Missing value analysis was conducted
- Data consistency across variables was verified
- All cleaning and transformation steps were documented to ensure reproducibility.

3. Data cleaning and preparation

Each CSV file was imported into Microsoft Excel and saved as an Excel Workbook.

• Data cleaning steps

Excluded variables

Some inconsistencies were identified between station names and geographic coordinates. Since geospatial analysis was not central to the business question, coordinate variables were excluded from the final analysis.

The columns:

- start_station_name
- end_station_name
- start_lat
- start_lng
- end_lat
- end_lng

contained a high number of missing and inconsistent values, particularly in later months of the dataset. These variables were excluded from the final analysis dataset.

- **Date and time formatting**

Using Power Query, the started_at and ended_at columns were standardized to the format dd.mm.yyyy hh:mm.

- **Data validation**

The following validation checks were performed:

- Verified that ride_id values were unique
- Checked for duplicate records
- Verified consistency of ride duration values
- Identified and removed invalid ride durations

- **New variables created**

Two new columns were added:

- ride_length – calculated as a difference between ended_at and started_at
- day_of_week – calculated from started_at column

- **Filtering invalid records**

Trips shorter than one minute were excluded from the analysis because they likely represented accidental unlocks, test rides, or system-related anomalies rather than meaningful customer behaviour.

The following records were removed:

Month	Rows Removed
May 2025	12.363
June 2025	19.605
July 2025	23.602
August 2025	22.917
September 2025	18.270
October 2025	16.991
November 2025	9.539
December 2025	3.955
January 2026	3.843
February 2026	4.019
March 2026	8.263
April 2026	12.414
Total	155.781

- **Combining the dataset**

All twelve monthly datasets were combined into a single dataset using Power Query. Because the combined dataset exceeded Excel worksheet row limitations, the data was loaded as:

- Connection Only
- Added to the Excel Data Model

This allowed the creation of Pivot Tables and visualizations using the complete dataset.

4. Data analysis

Three Pivot Tables were created to analyze:

- Average ride length by weekday and user type
- Number of rides by weekday and user type
- Number of rides by month and user type

The analysis focused on identifying behavioral differences between

- Casual riders
- Annual members

5. Visualizations and key findings

- **Average ride length by weekday**

The analysis shows that casual riders have significantly longer ride durations than annual members.

Casual riders

- Average ride duration during weekdays ranges between 19 and 23 minutes
- Ride duration increases considerably during weekends:
 - o Saturday: 26:13 minutes
 - o Sunday: 27:04 minutes

This pattern suggests that casual riders primarily use Cyclistic bikes for leisure and recreational activities.

Annual members

- Average ride duration remains relatively stable throughout the week
- Typical ride duration ranges between 12 and 14 minutes

This behaviour suggests that annual members primarily use bikes for commuting and short transportation trips.

- **Number of rides by weekday**

Although casual riders take longer rides, annual members complete a significant higher number of trips.

Key observations:

- Member rides peak during weekdays
- Casual rider activity increases during weekend
- On weekends, the difference between casual riders and annual members becomes much smaller.

Total rides during the analyzed period:

- Members: 3.583.755 rides
- Casual riders: 1.957.919 rides

These findings strongly suggest that annual members use Cyclistic bikes more frequently for daily commuting.

- **Number of rides by month**

A line chart was created to visualize monthly ride frequency for both user groups.

Key findings:

- Members consistently recorded more rides than casual riders throughout the year
- The highest ride numbers occurred between July and October
- Ride activity for both groups declined significantly during winter months (December and January)
- Ride frequency began increasing again in February and continued rising into spring.

The seasonal pattern suggests that weather conditions strongly influence bike usage.

6. Conclusions

The analysis identified clear behavioral differences between annual members and casual riders.

Annual members:

- Take shorter rides
- Ride more frequently
- Use bike consistently during weekdays
- Likely use Cyclistic bikes for commuting and transportation

Casual riders:

- Take longer rides
- Ride more frequently during weekends
- Show strong seasonal usage patterns
- Likely use bikes for leisure and recreation purposes

These differences can help Cyclistic design targeted marketing strategies aimed at converting casual riders into annual members.

7. Recommendations

Based on the analysis, the following recommendations are proposed:

- Promote cost savings for frequent riders: develop marketing campaigns emphasizing the long-term financial benefits of annual membership compared to single rides.
- Offer seasonal membership discounts: introduce discounted seasonal membership plans targeted at casual riders during high-activity months.
- Launch weekend conversion campaigns: since casual rider activity peaks during weekends, targeted weekend promotions could encourage casual riders to become annual members.